Pradeep Rajasekar

AI Research Engineer — 3D Computer Vision, On-Device Perception

ajishpradeep@gmail.com · Portfolio · GitHub · LinkedIn
New Taipei City, Taiwan · Open to relocation
English (professional) · Chinese (Basic)
EU Blue Card eligible

AI research engineer specialising in 3D human-motion perception and on-device inference. Architected the 2D-to-3D pose-lifting and club-tracking system behind a markerless motion-capture pipeline — engineering a frozen-pathway adapter that made regression on the general body model structurally impossible rather than merely unlikely, verified **bit-identical** to the base model by weight diff. Previously led the vision architecture for an open-set retail recognition system deployed across **7,000+ stores**, presented as a technical poster at **NVIDIA GTC 2025**.

CORE SKILLS

3D vision	3D human pose estimation · 2D-to-3D lifting · Multi-view geometry · Camera calibration · 3D reconstruction · Triangulation · Body tracking · 3D perception
On-device & edge	Apple Core ML · ARM / edge deployment · NVIDIA TensorRT · DeepStream · TAO Toolkit · Quantization & compression
Perception	Object detection (YOLO, DETR) · Open-set recognition · Metric learning / triplet loss · Embedding retrieval · Temporal consistency · Occlusion robustness
LLM systems	Agentic workflows & tool use · RAG · Structured grounding · Fine-tuning · Evaluation design
Engineering	Python · PyTorch · TensorFlow · JAX · HuggingFace · NumPy · GCP Vertex AI · JavaScript

EXPERIENCE

AI Research Engineer
May 2025 — Present

IdeasLab Formosa · Taipei, Taiwan

- Led R&D on the 2D-to-3D human pose lifting system behind **XView AI**, a markerless golf-swing analysis app: reduced mean per-joint 3D error **8 cm → 3 cm** (pelvis-relative) through temporal consistency modelling, motion-aligned lifting and spatial refinement, with **+30%** reconstruction fidelity under occlusion and fast motion.
- Shipped the full analysis pipeline **on-device** on Apple ARM silicon via Core ML — no cloud round-trip, no footage leaving the handset — in an iOS app used by **PGA Tour professionals**.
- Extended the pose model to club tracking without regressing the general body model: froze the backbone and every body keypoint channel and trained through a parallel adapter, leaving the deployed body pathway **bit-identical** to the base model (verified by weight diff) while club accuracy reached production quality.
- Diagnosed and corrected calibration and reconstruction faults across the multi-camera stack — **4.8×** lower reconstruction noise and **−73%** event-detection timing error — validated against published anthropometric ratios over **204** bone measurements.
- Architected a domain-grounded agentic LLM coaching system in which a deterministic rule engine computes every number and the model is constrained to narration, making fabricated biomechanics structurally impossible rather than merely unlikely; **69** weighted rules, **8** languages. Validated the rulebook against real sessions — rules firing on 100% of sessions were self-detecting bugs, and one intuitive explanation for a misfire was tested and disproved before it reached a user.
- Authored the technical proposal that won TAITRA's **"Go Healthy with Taiwan" 2025** award — **1 of 3 winners from 638 proposals across 55 countries**.

AI Engineer
Nov 2023 — Feb 2025

President Information Corp · Taipei, Taiwan

- Led the vision architecture for a real-time planogram compliance system — dense detection for localisation, a fine-tuned triplet-loss embedding space for identity — so adding a product costs vectors instead of a retraining run. Deployed across **7,000+ 7-ELEVEN stores** in Taiwan; the system's results (**99.23% / 98.93%** precision / recall on shelf detection) are documented in Ou et al., *Scientific Reports* (2025).
- Presented the methodology as a **technical poster at NVIDIA GTC 2025**.
- Deployed occlusion-robust handheld product recognition into Taiwan's **8th unmanned 7-Eleven store**, a **+30%** recognition gain under motion blur and arbitrary orientation.

- Built production inference on NVIDIA Metropolis microservices, TAO Toolkit, DeepStream and TensorRT across cloud and edge, in collaboration with NVIDIA.
- Built an LLM-assisted named-entity-recognition pipeline on GCP Document AI, and a predictive model over **5M+** data points at 95% confidence informing marketing and inventory decisions.

Software Developer — May 2017 — Aug 2021

AIBS Software Solutions · Coimbatore, India

- Built full-stack ERP systems for manufacturing workflows; engineered inventory and tax tracking that resolved **95%** of reported discrepancies.

RECOGNITION

"Go Healthy with Taiwan" — Winner — 2025

TAITRA · Taiwan Excellence

Sole author of the winning technical proposal; award to IdeasLab Formosa. **1 of 3 winners from 638 proposals across 55 countries**, USD 30,000.

Scalable Vision AI for Planogram Compliance — 2025

NVIDIA GTC 2025 — Technical Poster

Selected by technical review from an extended abstract. Research-to-production methodology for open-set retail product recognition.

Taiwan Expo Europe — engineering representative — 2026

TAITRA · EXPO XXI Warsaw

Selected to present the 3D motion-analysis work under the Taiwan Excellence banner.

RESEARCH

The Transformer Architecture — a mathematical walkthrough — 2024

Public technical writing · github.com/Ajishpradeep/Case_Study

One worked example carried from tokenisation and positional encoding through multi-head self-attention to the decoder's masked attention.

EDUCATION

MSc, Electrical Engineering & Computer Science — 2021 — 2023

National Taipei University of Technology · Taipei, Taiwan

GPA 3.8/4.0. Thesis: *Content and Spatial Aware Generative Model for Inpainting* — a GAN architecture combining contextual and spatial attention for low-data regimes.

BSc, Information Technology — 2011 — 2014

Sri Ramakrishna Mission Vidyalaya College · Coimbatore, India

GPA 7.9/10.

SELECTED PROJECTS

CarbonPass — 2026

Local-first vision-language model turning a factory's photographed paperwork into EU **CBAM** carbon accounting, with MILP production scheduling. Runs on-premise so documents never leave the building.
github.com/Ajishpradeep/CarbonPass

Magic Shuffle — 2026

A context-aware music recommender reading energy, sleep, stress, weather and calendar to choose and explain a track. Every suggestion is verified against Spotify's catalogue; a deterministic fallback works with **no API keys**.
github.com/Ajishpradeep/Magic-Shuffle

Data Automation Pipeline — 2025

A parallelised data-preparation pipeline converting PDFs, web pages and source files into clean Markdown for LLM consumption — preserving LaTeX formulae and code blocks, with optional OCR for scientific PDFs.
github.com/Ajishpradeep/data_automation_pipeline